

Math Hero Kids

Product Requirements Document (PRD)

Version: 1.0

1. Product Vision

Math Hero Kids is a gamified mathematics learning platform designed for children between 6 and 12 years old.

The objective is to transform daily math practice into a fun habit through progression systems, achievements, streaks, social interaction, friendly competition and long-term engagement.

The experience should feel like a combination of:

- Duolingo
- Pokémon
- Nintendo Games
- Khan Academy Kids

The goal is not only to teach multiplication, but to build a platform capable of supporting multiple math disciplines in the future.

2. Initial Scope

Version 1.0 will support:

✓ Multiplication

Future modules:

🚧 Division

🚧 Addition

🚧 Subtraction

🚧 Fractions

🚧 Decimals

🚧 Mental Math

The application architecture must be prepared for future expansion.

3. Target Audience

Primary Audience:

Children aged 6 to 12.

Secondary Audience:

Parents and guardians.

Future Audience:

Teachers and schools.

4. Core Principles

The application should:

- Encourage daily practice
- Reward consistency
- Reward effort
- Make mistakes feel safe
- Promote social learning
- Create long-term engagement

The application should never punish children harshly.

5. Family Profiles

A single device may be shared by multiple children.

Examples:

- Siblings sharing a tablet
- Family iPad
- Parent's phone

The application must support multiple child profiles.

6. Parent Account

Fields:

- Parent Name
- Parent Email
- Password

The parent account owns all child profiles.

7. Child Profile

Fields:

- Username
- Display Name
- Birth Date
- Avatar

Examples:

Lara Pedro Lucas

Each child has independent:

- Progress
 - XP
 - Achievements
 - Statistics
 - Friends
 - Challenges
-

8. Profile Selection Screen

Inspired by Netflix.

Shown immediately after opening the app.

Example:

Who is playing today?

 Lara

 Pedro

 Lucas

 New Profile

Switching profiles should require only one tap.

9. Daily Learning System

Children are encouraged to complete a daily challenge.

Daily completion grants:

- XP
- Trophy
- Streak Progress

The objective is consistency.

10. Progressive Difficulty

Children should not start with all multiplication tables.

Level 1:

Tables:

1 2 3

As levels increase:

New tables unlock.

Advanced ranges unlock.

11. Multiplication Settings

Configurable by parent.

Timer:

5s 10s 15s 20s 30s Unlimited

Default:

15s

Range:

1-10 1-12 1-15 1-20 Custom

12. Calendar System

Infinite scroll calendar.

Each day can be:

Future

Today

Completed

Failed

In Progress

Visual indicators:

🏆 Completed

💔 Failed

★ Perfect Week

13. XP System

Children earn XP for:

Correct answers

Completed blocks

Completed days

Completed weeks

Achievements

14. Levels

Examples:

Level 1: Math Explorer

Level 5: Math Adventurer

Level 10: Math Champion

Level 20: Math Master

Level 50: Math Legend

15. Achievements

Examples:

First Victory

7-Day Streak

30-Day Streak

100 Correct Answers

1000 Correct Answers

Perfect Week

Perfect Month

Speed Master

16. Friends System

Children may add friends using usernames.

Example:

@lara_menesca

@pedro_math

@joao123

Friend requests require acceptance.

17. Friend Challenges

Examples:

Complete Table 7 Today

Perfect Day Challenge

7-Day Streak Challenge

Results:

Victory

Tie

Incomplete

18. Weekly Clubs

Friends may join clubs.

Example:

Math Warriors

Members work together toward goals.

Example Goal:

Complete 100 tables this week.

Rewards:

Badges

Trophies

Club Achievements

19. Dashboard

Main screen contains:

- Current Level
- XP
- Current Streak
- Weekly Progress
- Daily Progress
- Trophy Count
- Friend Activity

Mascot Milo appears here.

20. Internationalization

Supported Languages:

English

Portuguese

Spanish

French

All application text must come from translation files.

21. Offline Mode

Guest Mode:

Local Storage

Logged Users:

Cloud Synchronization

The application should continue working without internet whenever possible.

22. Future Monetization

Potential premium features:

Custom Avatars

Special Themes

Advanced Statistics

Parent Reports

School Mode

No pay-to-win mechanics.

23. Success Metrics

Daily Active Users

Weekly Active Users

Average Streak Length

Monthly Retention

Completed Challenges

Friend Challenges Created

Club Participation

24. Long-Term Vision

Become the most engaging mathematics learning platform for children, combining educational effectiveness with game-quality engagement.